

Supplementary File 9. Domain-Specific Evidence Tables

Purpose

This supplementary file presents the charted findings from included sources by evidence domain. It has been revised to match the source-level domain coding matrix in Supplementary File 6.

Because several sources contributed to more than one domain, some sources appear in more than one table. However, each source appears only in domains to which it was explicitly coded in Supplementary File 6.

Reconciled Domain Counts

Domain code	Evidence domain	Number of sources
D1	Biological interaction between malaria and red blood cell polymorphisms	20
D2	Spatial co-risk and malaria-HbS/SCD geography	10
D3	Clinical convergence and under-recognition	10
D4	Service-mediated visibility and routine data	11
D5	Structural vulnerability and access to essential SCD services	20
D6	Programme response and screening-to-care systems	19
D7	Use of syndemic theory in malaria-SCD literature	6

Total domain assignments		96
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The total number of included sources remains **68**. The total number of domain assignments is greater than 68 because some sources were coded into more than one domain.

Table 1. Domain 1 Evidence: Biological Interaction Between Malaria and Red Blood Cell Polymorphisms

Study ID	Source	Country or region	Red cell trait or biological focus	Main charted finding	Contribution to review
S001	Allison, 1954	East Africa	HbAS	Sickle cell trait was associated with protection against malaria in endemic African populations.	Foundational evidence for malaria-HbS biological interaction.
S002	Allison, 1954	East Africa and broader African relevance	HbS distribution	Sickle cell trait distribution was geographically linked to malaria-endemic areas.	Connected biological selection to spatial malaria ecology.

S003	Aidoo et al., 2002	Kenya	HbAS	The sickle cell gene was associated with protection against malaria morbidity and mortality.	Strengthened epidemiological evidence for HbAS protection.
S004	Williams et al., 2005	Kenya	HbAS	HbAS was associated with reduced risk of severe <i>P. falciparum</i> malaria and malaria-related childhood outcomes.	Provided core evidence for phenotype-specific HbAS protection.
S005	Williams et al., 2005	Kenya	HbAS and alpha-thalassaemia	Co-inheritance of HbAS and alpha-thalassaemia modified malaria-protective effects.	Showed epistasis and limits of single-polymorphism models.

S006	Taylor et al., 2012	Global with SSA relevance	Haemoglobinopathies	Haemoglobin variants modify malaria susceptibility and severity.	Synthesised evidence on malaria and haemoglobin variants.
S007	Taylor et al., 2013	Global with SSA relevance	HbAS mechanisms	Reviewed mechanisms proposed to explain HbAS protection against malaria.	Supported mechanistic interpretation of HbAS protection.
S008	Piel et al., 2010	Global with strong Africa relevance	HbS and malaria endemicity	HbS allele frequency showed broad geographic correspondence with historical malaria endemicity.	Linked malaria selection to HbS persistence.
S011	Modiano et al., 2001	Burkina Faso	HbC	Haemoglobin C was associated with protection against clinical <i>P.</i>	Extended malaria-protective haemoglobin variation beyond HbS.

				<i>falciparum</i> malaria.	
S012	Mockenhaupt et al., 2004	Ghana	HbC and HbAS	Haemoglobin variants were associated with resistance to severe malaria.	Added West African evidence on haemoglobin variants and severe malaria.
S013	Mockenhaupt et al., 2004	Ghana	Alpha-thalassaemia	Alpha-plus-thalassaemia protected African children against severe malaria.	Added evidence for non-HbS red cell protection.
S014	Williams et al., 2005	Kenya	Alpha-thalassaemia	Heterozygous and homozygous alpha-plus-thalassaemias protected against severe and fatal malaria.	Supported phenotype-specific malaria protection by alpha-thalassaemia.
S015	Ruwende et al., 1995	Africa, multicountry relevance	G6PD deficiency	G6PD deficiency was consistent	Added G6PD to the malaria-selected red

				with selection by resistance to severe malaria.	cell polymorphism landscape.
S016	Mbanefo et al., 2017	Global with African relevance	G6PD deficiency	Evidence on G6PD deficiency and malaria was heterogeneous across settings and outcomes.	Supported context-dependent interpretation of malaria-red cell relationships.
S017	Rowe et al., 2007	Mali and Kenya	ABO blood group	Blood group O reduced rosetting and protected against severe malaria.	Identified rosetting as a mechanism of malaria severity modification.
S018	Fry et al., 2008	Gambia, Kenya, Malawi	ABO genetic variation	ABO genetic variation was associated with susceptibility to severe malaria.	Reinforced ABO as malaria-relevant red cell biology.
S019	Kariuki et al., 2020	Kenya	Dantu	Dantu-associated red cell	Added erythrocyte biomechanics

				membrane tension reduced parasite invasion and protected against severe malaria.	to the biological interaction domain.
S020	MalariaGEN, 2014	Multicountry Africa and Asia	Host resistance loci	Severe malaria susceptibility was shaped by multiple host genetic loci.	Supported multi-locus interpretation of malaria resistance.
S021	Weatherall, 2010	Global with Africa relevance	Red cell variation	Reviewed red cell genetic variation shaped by malaria selection.	Provided broader biological context for malaria-driven erythrocyte adaptation.
S062	Wonkam et al., 2020	Africa	SCD genomics and public health	Linked African SCD genomics to diagnosis, burden, and	Connected inherited risk to public health and care-system implications.

				public health gaps.	
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Table 2. Domain 2 Evidence: Spatial Co-Risk and Malaria-HbS/SCD Geography

Study ID	Source	Country or region	Spatial focus	Main charted finding	Contribution to review
S002	Allison, 1954	East Africa and broader African relevance	Distribution of sickle trait	Sickle cell trait distribution corresponded with malaria-endemic areas.	Early spatial support for the malaria selection hypothesis.
S008	Piel et al., 2010	Global with strong Africa relevance	HbS allele frequency and malaria endemicity	HbS distribution showed geographic correspondence with historical malaria endemicity.	Linked inherited sickle-related risk to malaria ecology.
S009	Piel et al., 2013	Global with strong SSA relevance	Neonatal HbS and HbSS births	HbS frequency and HbSS births were geographically	Supported spatial planning for SCA burden.

				heterogeneous, with high burden in SSA.	
S010	Piel et al., 2013	Global with strong SSA relevance	SCA under-five burden	Estimated SCA burden among children under five was concentrated in high-burden African settings.	Linked spatial burden to survival and service need.
S022	Bhatt et al., 2015	Endemic Africa	Malaria burden and control	Malaria burden declined with control but remained spatially heterogeneous.	Established dynamic malaria exposure geography.
S023	Gething et al., 2011	Global with strong Africa relevance	<i>P. falciparum</i> endemicity	<i>P. falciparum</i> endemicity varied substantially at subnational scale.	Provided malaria exposure geography for co-risk interpretation.

S024	Weiss et al., 2019	Global with strong SSA relevance	Malaria prevalence, incidence, and mortality	Malaria burden remained geographically heterogeneous from 2000 to 2017.	Provided contemporary malaria burden mapping.
S025	Dalrymple et al., 2015	Africa relevance	Routine facility malaria data	Facility data could inform malaria mapping when interpreted through care-seeking and reporting processes.	Demonstrated service-mediated spatial burden estimation.
S026	Alegana et al., 2016	Kenya	Facility catchments and malaria incidence	Routine facility data could estimate malaria incidence when adjusted for treatment seeking and catchment effects.	Supported catchment-sensitive interpretation of routine disease signals.

S027	Ndeezi et al., 2016	Uganda	National sickle surveillance	Sickle cell trait and SCD prevalence varied across regions.	Demonstrated subnational sickle-related risk heterogeneity.
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Table 3. Domain 3 Evidence: Clinical Convergence and Under-Recognition

Study ID	Source	Country or region	Clinical focus	Main charted finding	Contribution to review
S030	Kato et al., 2018	Global	SCD pathophysiology	SCD involves haemolysis, vaso-occlusion, anaemia, and organ injury.	Provided pathophysiological basis for clinical vulnerability.
S031	White, 2018	Global with malaria-endemic relevance	Malaria-related anaemia	Malaria causes anaemia through red cell destruction, inflammation, splenic clearance, and impaired erythropoiesis.	Supported severe anaemia as a malaria-SCD convergence syndrome.

S032	Marsh et al., 1995	Kenya	Severe malaria phenotypes	Life-threatening malaria in African children includes distinct clinical phenotypes.	Supported phenotype-specific interpretation of severe malaria.
S033	Maitland et al., 2019	Uganda and Malawi	Severe anaemia and transfusion timing	Severe anaemia was an acute paediatric syndrome requiring transfusion-relevant care.	Linked clinical convergence to acute-care capacity.
S034	Maitland et al., 2019	Uganda and Malawi	Severe anaemia and transfusion volume	Transfusion volume strategy was clinically important in severe anaemia management.	Supported transfusion readiness as a care requirement.
S035	Uyoga et al., 2022	Uganda	Severe anaemia, SCA, and malaria biomarkers	Previously unknown SCA was identified among	Core evidence for clinical convergence

				children admitted with severe anaemia; malaria-positive SCA cases had lower parasite density and PfHRP2 than HbAA children.	and under-recognition.
S036	Awotua-Efebo et al., 2004	Nigeria	Malaria parasitaemia in SCA	Malaria parasite density was assessed among children with SCA.	Added clinical evidence on malaria positivity in SCA.
S037	Nsagha et al., 2020	Cameroon	Malaria and SCD in children	Reported malaria prevalence, parasite density, and clinical patterns among children with SCD.	Added hospital evidence on malaria-SCD overlap.

S038	Sumbele et al., 2016	Cameroon	Malaria and moderate-to-severe anaemia	Malaria was associated with moderate-to-severe anaemia among children.	Supported malaria-anaemia convergence.
S040	Masamu et al., 2024	Tanzania	Very severe anaemia in SCD	Very severe anaemia was an important adverse clinical event among people with SCD.	Linked severe anaemia to SCD outcomes in SSA.

Table 4. Domain 4 Evidence: Service-Mediated Visibility and Routine Data

Study ID	Source	Country or region	Visibility mechanism	Main charted finding	Contribution to review
S025	Dalrymple et al., 2015	Africa relevance	Routine facility data	Routine malaria data require interpretation through reporting and care-seeking processes.	Provided methodological analogy for service-visible disease burden.

S026	Alegana et al., 2016	Kenya	Facility data and catchments	Routine malaria incidence estimates depended on treatment seeking and catchment assumptions.	Supported interpretation of routine data as service-mediated signals.
S027	Ndeezi et al., 2016	Uganda	Screening-linked surveillance	Dried blood spot testing revealed regional variation in sickle trait and SCD prevalence.	Showed how screening makes hidden burden visible.
S028	Grosse et al., 2011	Africa	Under-recognised mortality	SCD was described as a neglected cause of early childhood mortality in Africa.	Supported hidden burden and missed diagnosis framing.
S029	GBD 2021 SCD Collaborators, 2023	Global with strong SSA relevance	Mortality attribution	Total SCD-attributable mortality exceeded cause-	Demonstrated under-attribution of SCD mortality.

				specific SCD mortality.	
S035	Uyoga et al., 2022	Uganda	Hospital genotyping	Previously unknown SCA was identified among children admitted with severe anaemia.	Showed that hospital admission does not guarantee diagnostic visibility.
S047	Tshilolo et al., 2009	Democratic Republic of Congo	Neonatal screening	Neonatal screening identified SCA in a Central African setting.	Demonstrated screening visibility.
S048	Rahimy et al., 2009	Benin	Newborn screening and follow-up	Screening was linked to enrolment and follow-up.	Showed service visibility through screening-to-care linkage.
S049	Odunvbun et al., 2008	Nigeria	Hospital newborn screening	Hospital-based screening identified SCD among newborns.	Added evidence that facility-based screening can reveal burden.

S051	Ohene-Frempong et al., 2008	Ghana	Screening, tracking, and clinic enrolment	Newborn screening included tracking, result return, and clinic enrolment.	Showed visibility requires follow-up systems, not testing alone.
S059	SickleInAfrica Consortium, 2020	Multicountry Africa	Registry and data systems	African SCD registry and research infrastructure was being developed.	Supported routine visibility through data systems and registries.

Table 5. Domain 5 Evidence: Structural Vulnerability and Access to Essential SCD Services

Study ID	Source	Country or region	Care or vulnerability focus	Main charted finding	Contribution to review
S010	Piel et al., 2013	Global with strong SSA relevance	SCA burden and survival	SCA births were linked to child survival and health-system need.	Connected inherited burden to service need.
S028	Grosse et al., 2011	Africa	Early childhood mortality	SCD was under-recognised as a cause of early	Linked neglect and delayed diagnosis to

				childhood mortality.	mortality vulnerability.
S029	GBD 2021 SCD Collaborators, 2023	Global with strong SSA relevance	Mortality attribution	Total SCD-attributable mortality exceeded cause-specific SCD deaths.	Showed the mortality consequences of hidden burden.
S030	Kato et al., 2018	Global	SCD pathophysiology	SCD causes chronic anaemia, organ injury, and acute complications.	Explained biological vulnerability requiring continuous care.
S033	Maitland et al., 2019	Uganda and Malawi	Transfusion timing	Severe anaemia required transfusion-relevant acute care.	Linked clinical vulnerability to transfusion readiness.
S034	Maitland et al., 2019	Uganda and Malawi	Transfusion volume	Transfusion strategy affected severe anaemia management.	Supported the need for safe, responsive transfusion systems.
S039	Makani et al., 2011	Tanzania	SCA mortality	Mortality among	Demonstrated survival

				people with SCA remained substantial in an African cohort.	vulnerability in African SCD care settings.
S040	Masamu et al., 2024	Tanzania	Very severe anaemia in SCD	Very severe anaemia was associated with adverse SCD outcomes.	Linked SCD morbidity to acute care needs.
S041	Gaston et al., 1986	United States, foundational care evidence	Penicillin prophylaxis	Oral penicillin reduced pneumococcal septicemia among young children with SCA.	Provided foundational evidence for early preventive care.
S042	Quinn et al., 2010	United States, comparative benchmark	Organised care and survival	Newborn screening and organised care were associated with improved survival.	Provided benchmark for expected benefit of early diagnosis and care.
S043	WHO, 2025	Global	Essential care	Early diagnosis,	Supported essential

				vaccination, hydroxyurea, transfusion, and comprehensive care are central to SCD management.	service package.
S044	Tshilolo et al., 2019	Angola, DRC, Kenya, Uganda	Hydroxyurea	Hydroxyurea was feasible, safe, and clinically beneficial for children with SCA in SSA.	Provided SSA evidence for disease-modifying therapy.
S045	Opoka et al., 2017	Uganda	Hydroxyurea in malaria-endemic setting	Hydroxyurea use was evaluated in a malaria-endemic African region.	Addressed safety concerns around hydroxyurea in malaria-exposed children.
S046	John et al., 2020	Uganda	Hydroxyurea dose escalation	Hydroxyurea dose escalation was evaluated among	Added evidence for optimising hydroxyurea delivery in SSA.

				children with SCA.	
S048	Rahimy et al., 2009	Benin	Screening and follow-up	Screening was linked to enrolment and follow-up.	Demonstrated that screening benefit depends on care linkage.
S052	Therrell et al., 2015	Africa	Screening implementation barriers	Newborn screening initiatives faced care linkage and implementation barriers.	Identified structural barriers to screening benefit.
S055	Nnodu et al., 2024	Africa	Screening initiatives	Screening programmes faced barriers in policy, enrolment, linkage, and sustainability.	Highlighted structural gaps across the screening-to-care cascade.
S060	Makani et al., 2013	Africa	SCD care systems	African SCD care required stronger diagnosis, comprehensive care, and research	Supported structural vulnerability and care-system strengthening.

				infrastructure .	
S061	Ware et al., 2017	Global with SSA relevance	SCD management	SCD management requires prevention, acute care, disease-modifying therapy, and long-term care.	Provided broader clinical management framework.
S062	Wonkam et al., 2020	Africa	Genomics and public health	African SCD genomics and public health systems require stronger diagnosis and care infrastructure .	Linked inherited burden to health-system response.

Table 6. Domain 6 Evidence: Programme Response and Screening-to-Care Systems

Study ID	Source	Country or region	Programme focus	Main charted finding	Contribution to review
S027	Ndeezi et al., 2016	Uganda	Sickle surveillance	Screening-linked surveillance	Demonstrated surveillance as a

			using dried blood spots	revealed spatial variation in sickle trait and SCD prevalence.	programme-planning tool.
S043	WHO, 2025	Global	Essential care guidance	SCD care includes diagnosis, vaccination, hydroxyurea, transfusion, and comprehensive management.	Supported programme content for essential care.
S044	Tshilolo et al., 2019	Angola, DRC, Kenya, Uganda	Hydroxyurea delivery	Hydroxyurea was feasible and beneficial for children with SCA in SSA.	Supported integration of disease-modifying therapy into SCD programmes.
S045	Opoka et al., 2017	Uganda	Hydroxyurea in malaria-endemic setting	Hydroxyurea use was evaluated in children with SCA in a malaria-	Supported programme decisions on hydroxyurea safety in malaria-

				endemic region.	endemic settings.
S046	John et al., 2020	Uganda	Hydroxyurea dose escalation	Dose escalation of hydroxyurea was evaluated among children with SCA.	Supported optimisation of hydroxyurea delivery.
S047	Tshilolo et al., 2009	Democratic Republic of Congo	Neonatal screening	Neonatal screening for SCA was feasible in DRC.	Added Central African screening evidence.
S048	Rahimy et al., 2009	Benin	Screening and follow-up	Newborn screening was linked to enrolment and follow-up.	Demonstrated early screening-to-care linkage.
S049	Odunvbun et al., 2008	Nigeria	Hospital-based newborn screening	Hospital newborn screening identified SCD.	Added West African screening feasibility evidence.
S050	Nnodu et al., acceptability of newborn screening	Nigeria	Screening acceptability	Newborn screening for SCD was acceptable in	Supported social feasibility of screening.

				the Nigerian study context.	
S051	Ohene-Frempong et al., 2008	Ghana	Newborn screening programme	Screening included tracking, result return, and clinic enrolment.	Provided a leading African screening-to-care model.
S052	Therrell et al., 2015	Africa	Newborn screening implementation	African newborn screening initiatives faced implementation and linkage barriers.	Identified programme constraints.
S053	McGann, 2014	Africa	Screening priority	Newborn screening was identified as a public health priority for SCD in Africa.	Supported policy rationale for screening expansion.
S054	Green et al., 2022	Multicountry SSA	CONSA	CONSA linked newborn	Provided an integrated multicountry

				screening to standardised care and mortality ascertainment .	implementation model.
S055	Nnodu et al., 2024	Africa	Screening initiatives	African newborn screening initiatives faced gaps in policy, enrolment, linkage, and sustainability.	Synthesised current programme evidence and gaps.
S056	Okeke et al., 2024	Resource-poor settings with Africa relevance	Screening sustainability	Screening sustainability was constrained by financing, policy, workforce, laboratory, and follow-up barriers.	Highlighted sustainability requirements.
S057	Steele et al., 2019	Ghana, Martinique, United States	Point-of-care testing	HemoTypeS C supported SCD screening in	Added diagnostic innovation evidence.

				low-resource settings.	
S058	Nnodu et al., 2019	Nigeria	HemoTypeS C implementation	Field use of HemoTypeS C showed practical challenges in decentralised diagnosis.	Linked point-of-care testing to implementation conditions.
S059	SickleInAfrica Consortium, 2020	Multicountry Africa	Registry and network infrastructure	SickleInAfrica developed registry and research infrastructure .	Supported programme data systems and networked implementation.
S060	Makani et al., 2013	Africa	SCD care systems	African SCD care required stronger diagnosis, comprehensive care, and research infrastructure .	Supported programme strengthening and system-building.

Table 7. Domain 7 Evidence: Use of Syndemic Theory in Malaria-SCD Literature

Study ID	Source	Country or region	Theory or methodological focus	Main charted finding	Contribution to review
S063	Singer and Clair, 2003	Global theoretical relevance	Syndemic theory	Syndemics were defined as disease clustering and interaction in biosocial context.	Provided conceptual foundation for syndemic framing.
S064	Singer et al., 2017	Global theoretical relevance	Biosocial syndemic theory	Syndemics were framed as interactions between diseases and structural conditions.	Supported multilevel interpretation of malaria-SCD evidence.
S065	Tsai et al., 2017	Global methodological relevance	Syndemic measurement critique	Co-occurrence alone was insufficient to establish syndemic interaction.	Supported cautious evidence classification.
S066	Tsai, 2018	Global methodological relevance	Syndemic theory and evidence	Syndemic claims require	Informed the distinction between co-

				analytic specificity about interaction and excess burden.	occurrence and stronger syndemic evidence.
S067	Mendenhall, 2017	Global theoretical relevance	Syndemics and global health	Syndemic theory was positioned as a framework for global health research.	Supported use of syndemic theory in SSA health-system interpretation .
S068	Mendenhall, 2021	Global theoretical relevance	Syndemic application and critique	Syndemic reasoning should be applied carefully in clinical and global health research.	Supported bounded use of syndemic theory in the review.

Table 8. Cross-Domain Evidence Configuration

Evidence segment	Main mapped result	Main supporting sources
Malaria-HbS biological interaction	HbAS protects against severe <i>P. falciparum</i> malaria.	S001, S003, S004
HbS persistence and SCA burden	HbS frequency and HbSS births are spatially	S008, S009, S010

	heterogeneous and concentrated in many malaria-endemic African settings.	
Wider red cell adaptation	HbC, alpha-thalassaemia, G6PD deficiency, ABO, Dantu, and epistasis modify malaria susceptibility or severity.	S011 to S020
Malaria spatial heterogeneity	Malaria prevalence, incidence, and mortality vary geographically and over time.	S022, S023, S024
Spatial co-risk and planning	HbS/SCA burden and malaria exposure are spatially patterned, but few sources link both to facility readiness or referral systems.	S008, S009, S022, S024, S027
Clinical convergence	Severe anaemia is the main clinical bridge between malaria and SCA.	S030, S031, S033, S034, S035
Under-recognition	SCA may be hidden among children admitted with severe anaemia.	S035
Service-mediated visibility	SCD burden becomes visible through screening, hospital genotyping, registries, and mortality attribution.	S027, S029, S035, S051, S059
Structural vulnerability	Access to prophylaxis, hydroxyurea, transfusion,	S039, S041, S043, S044, S055

	referral, and follow-up shapes morbidity and survival.	
Programme response	Screening-to-care systems are required to convert diagnosis into clinical benefit.	S047 to S060
Syndemic theory	Co-occurrence alone is insufficient; stronger evidence requires interaction, structural context, or pathway integration.	S063 to S068